

The Databricks Data Intelligence Platform

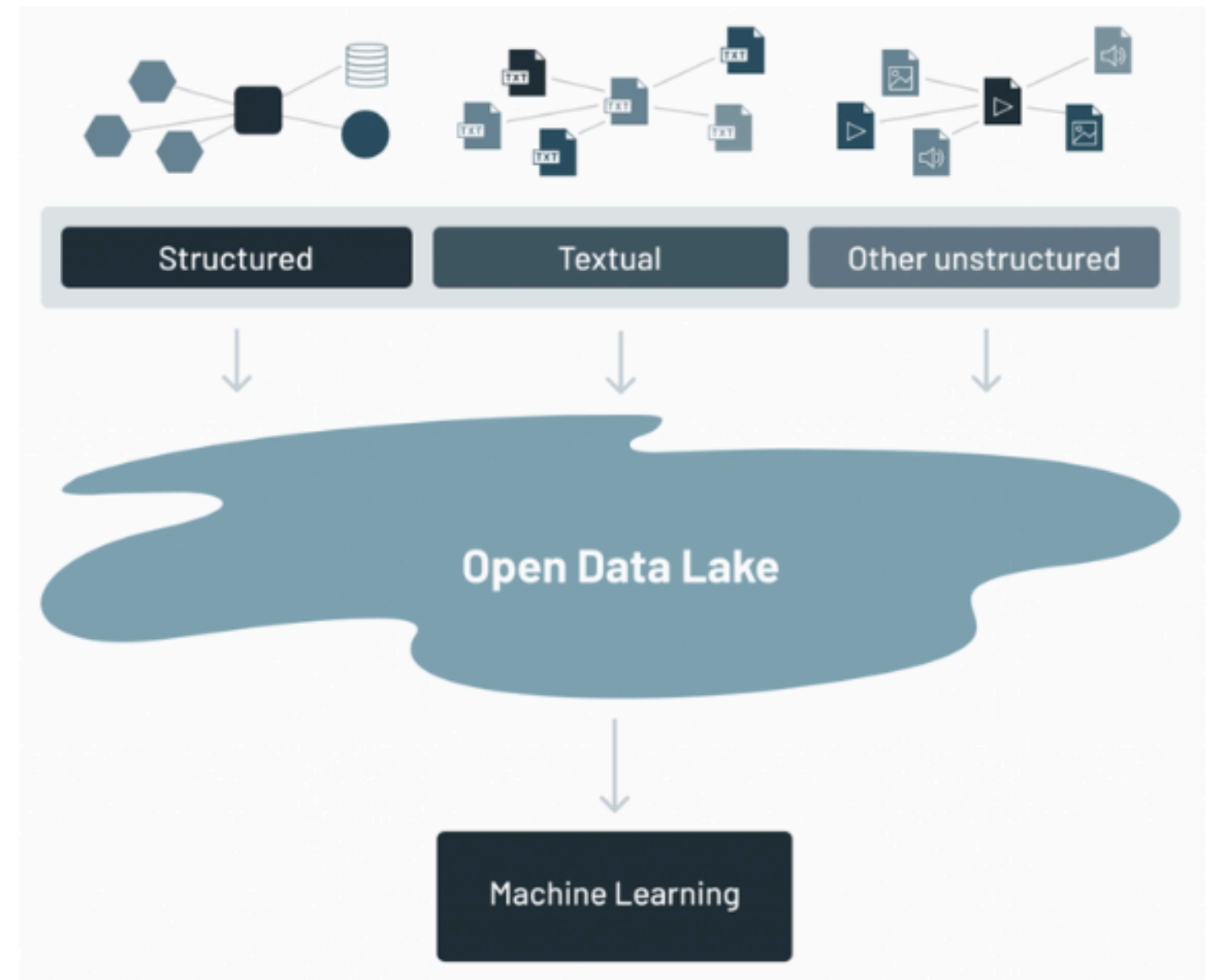
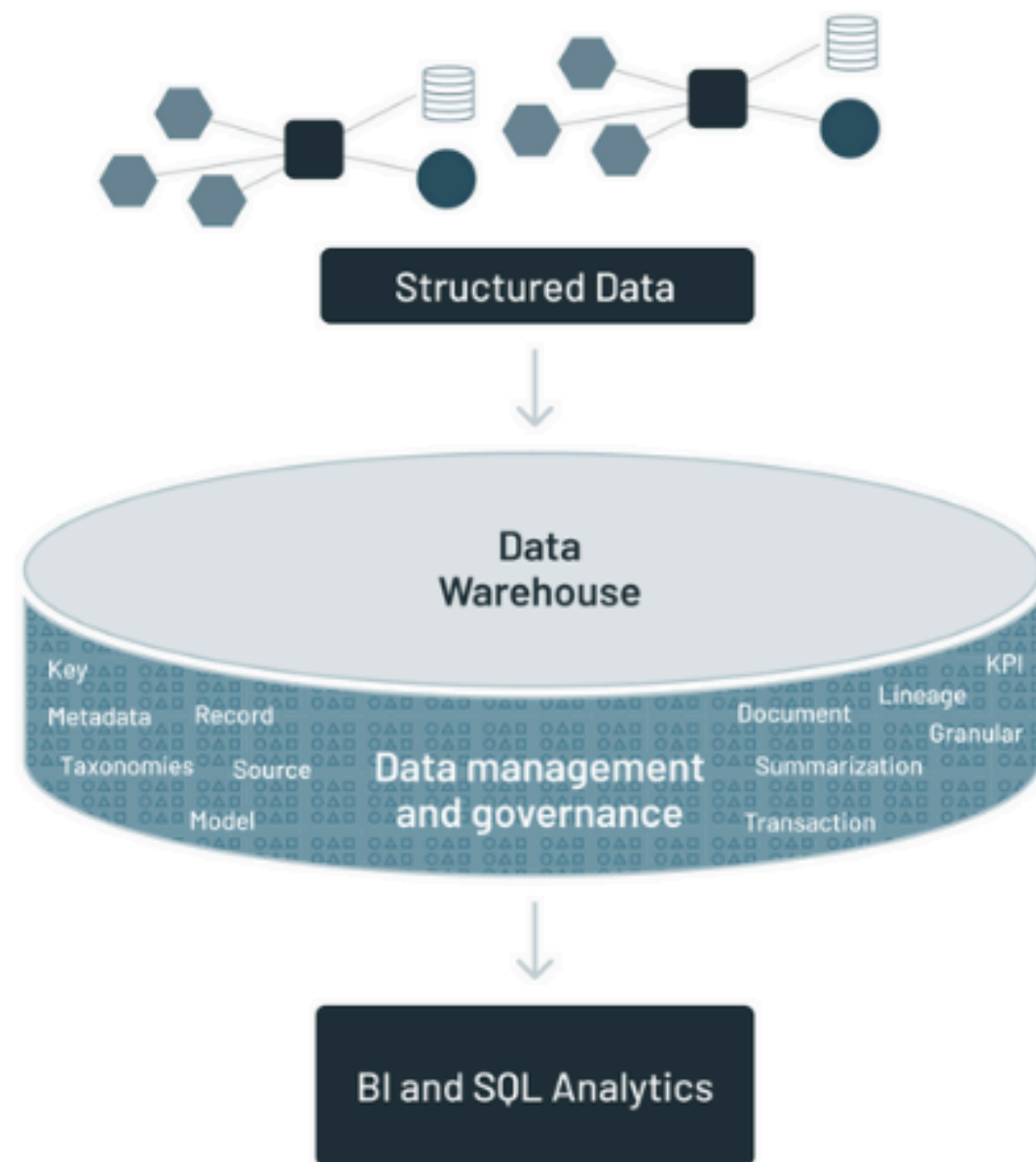
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Kevin Barlow

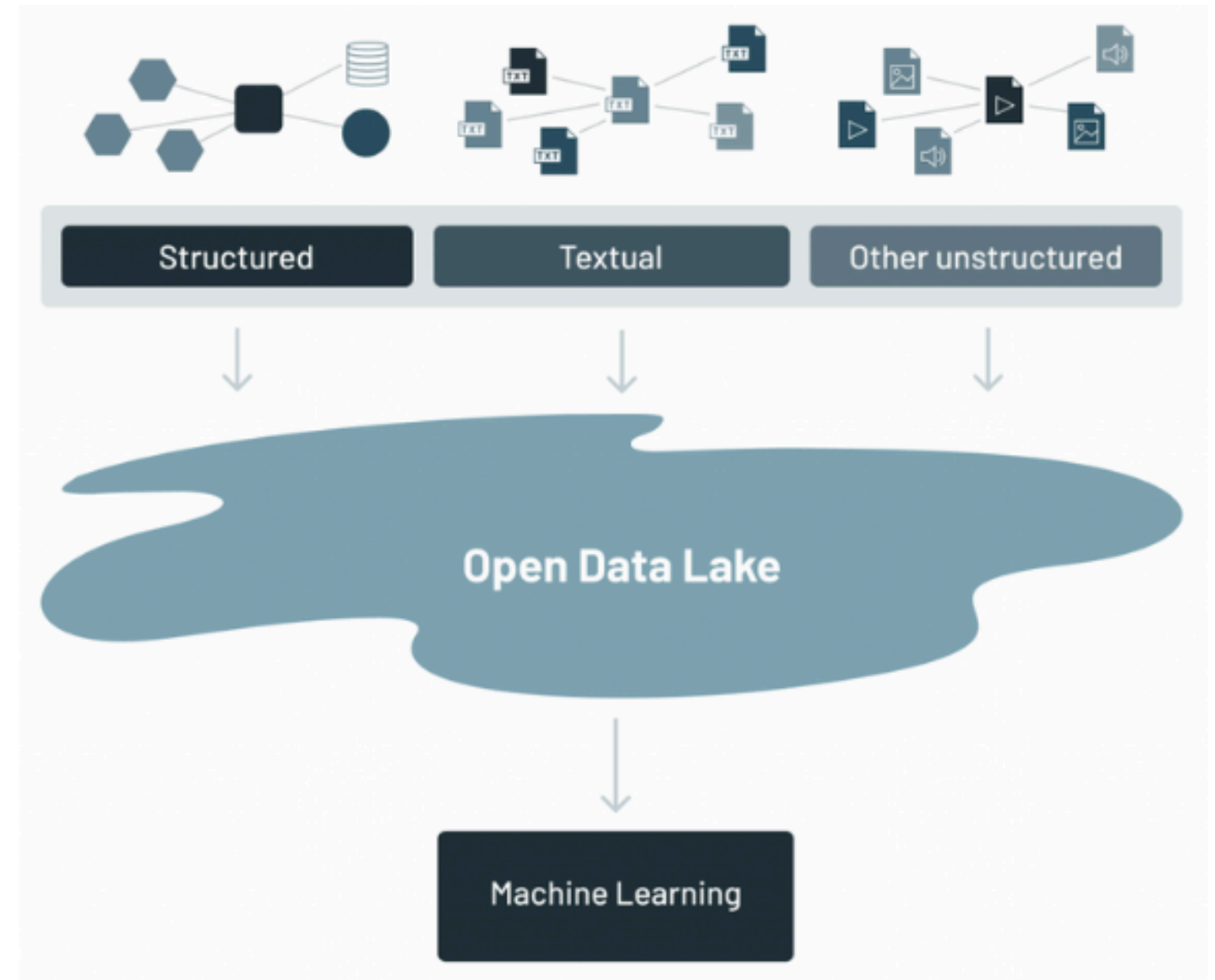
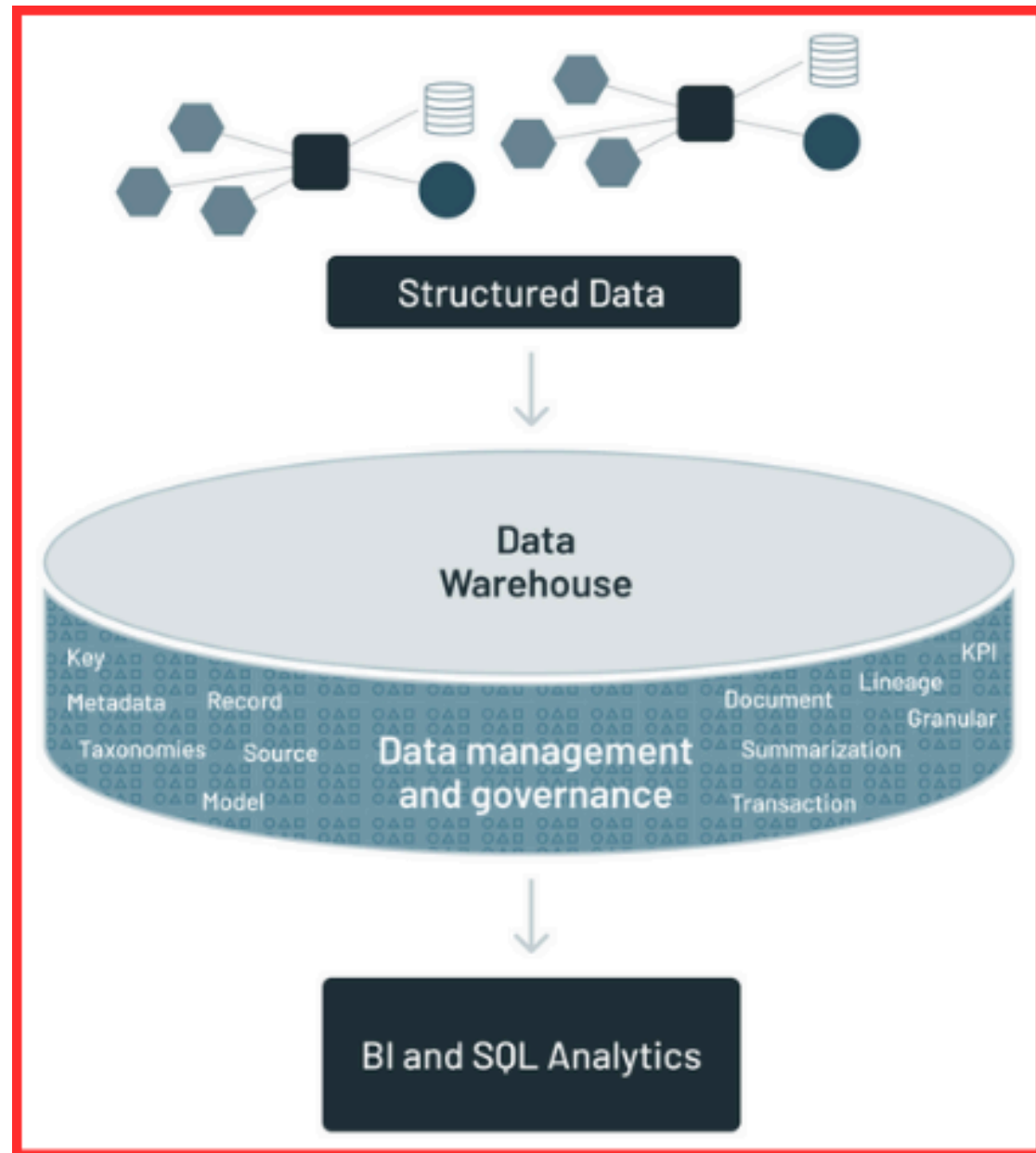
Data Analytics Practitioner

Architecture Options



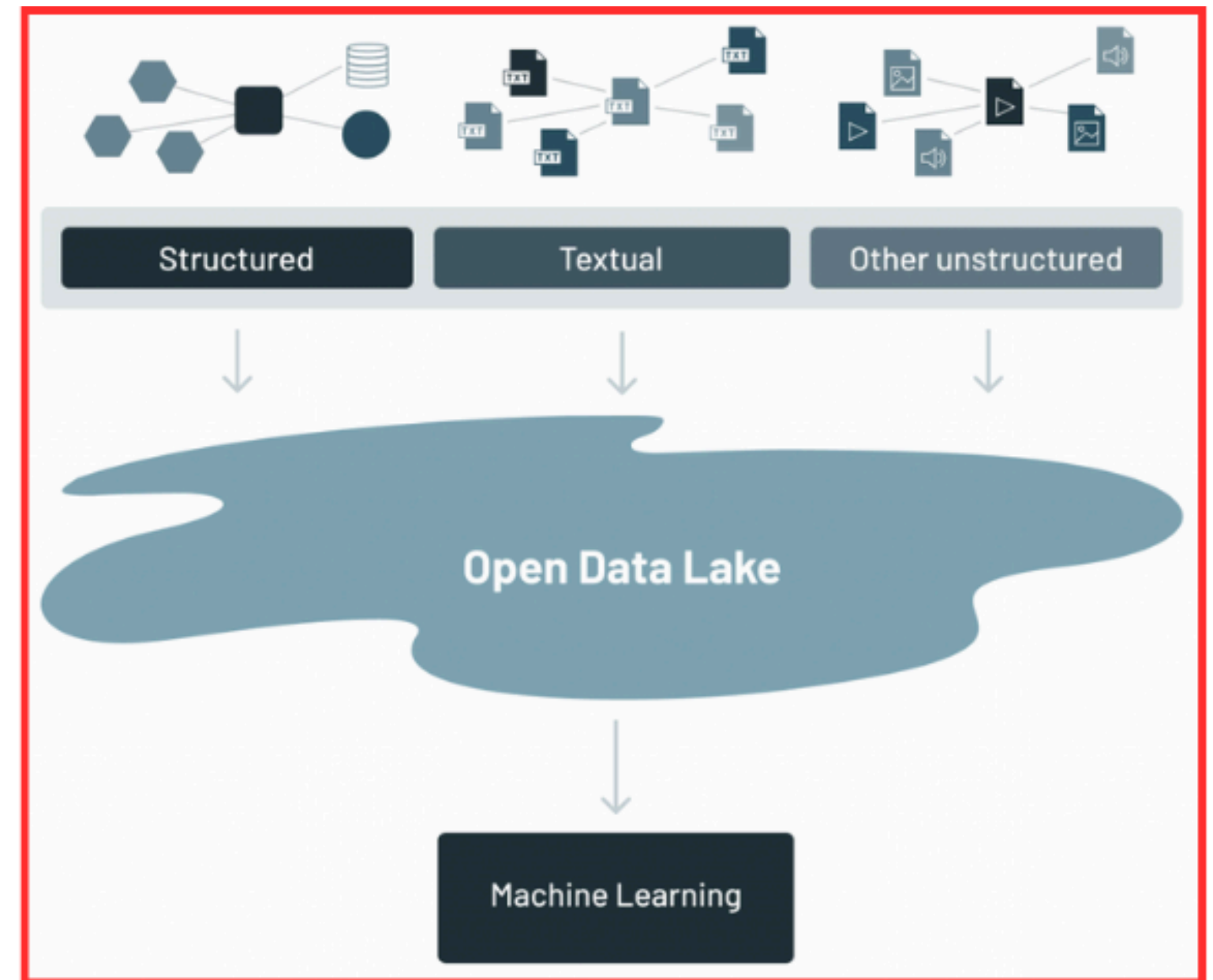
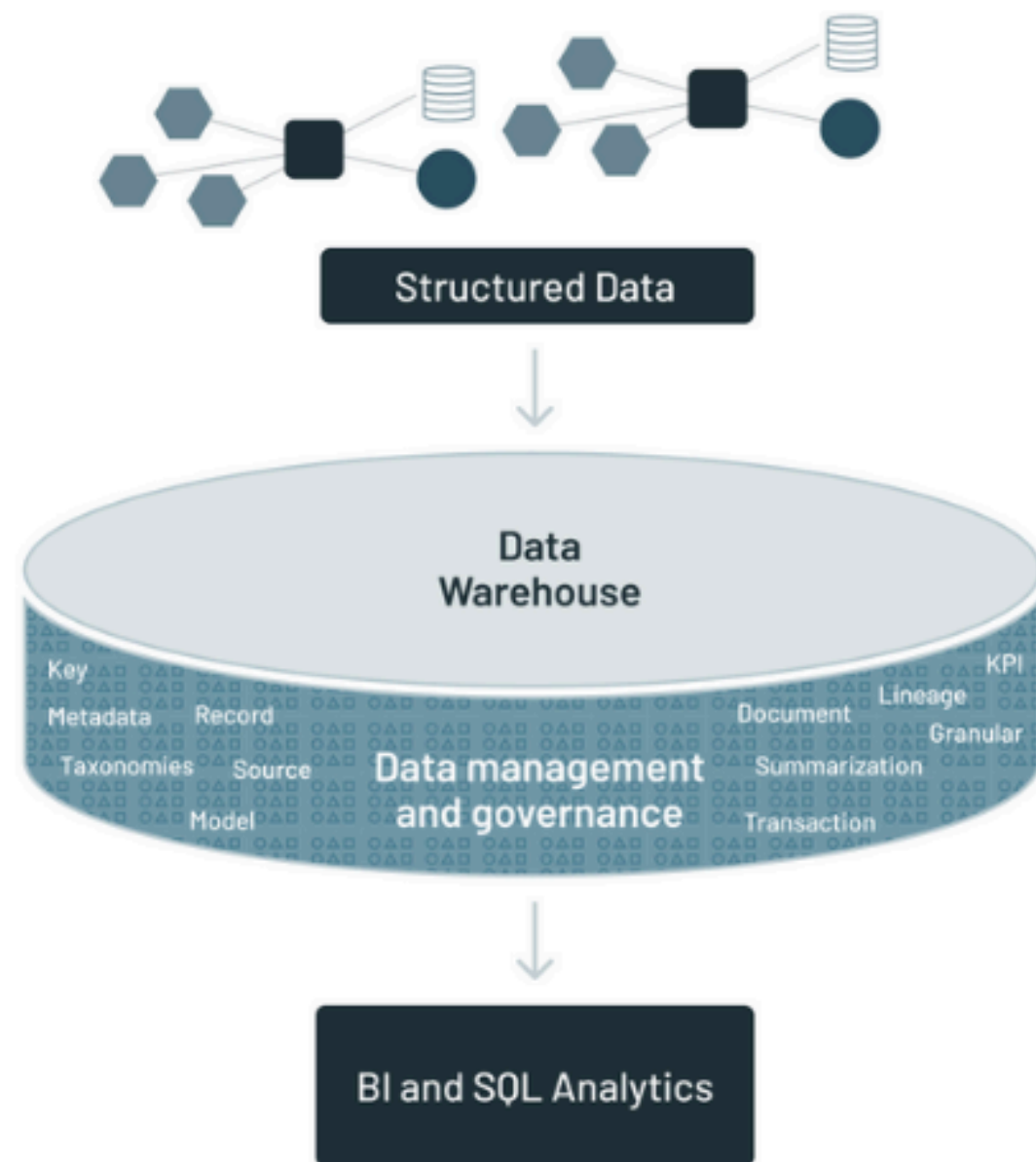
¹ <https://www.databricks.com/blog/2021/05/19/evolution-to-the-data-lakehouse.html>

Birth of the Lakehouse



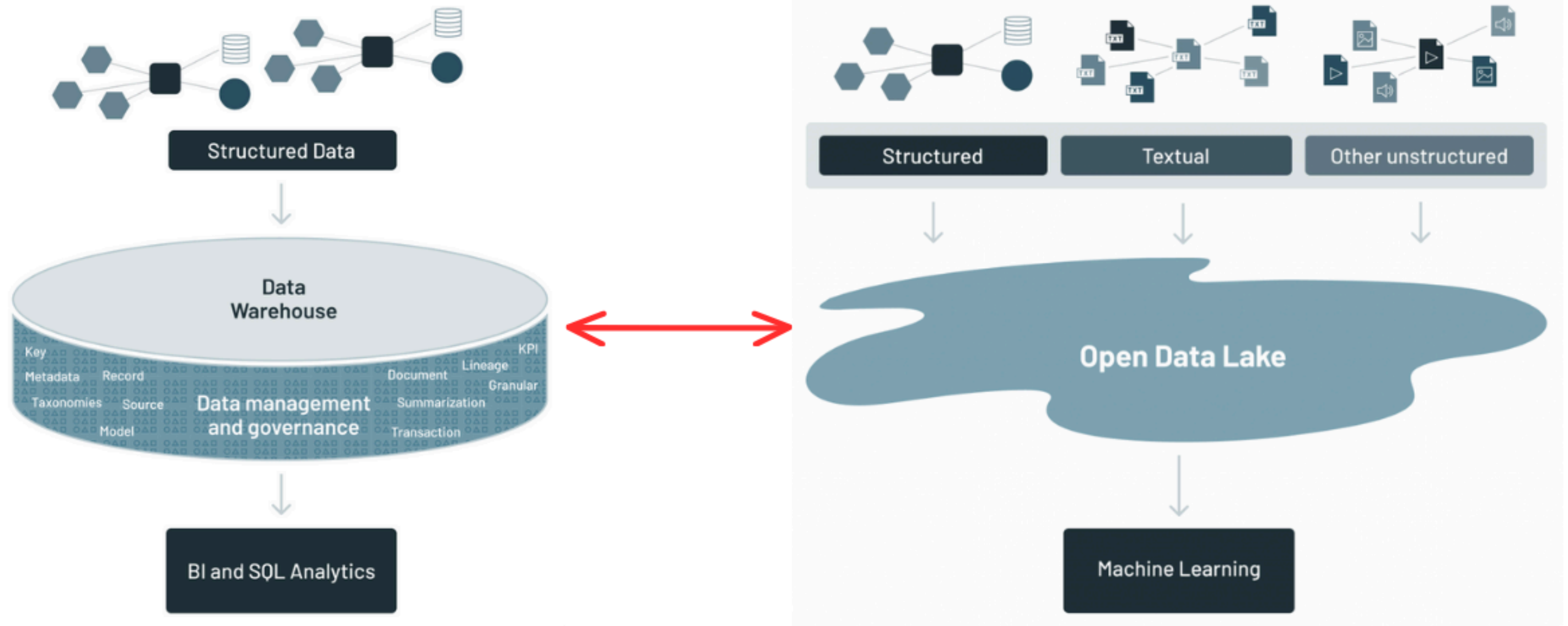
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Birth of the Lakehouse



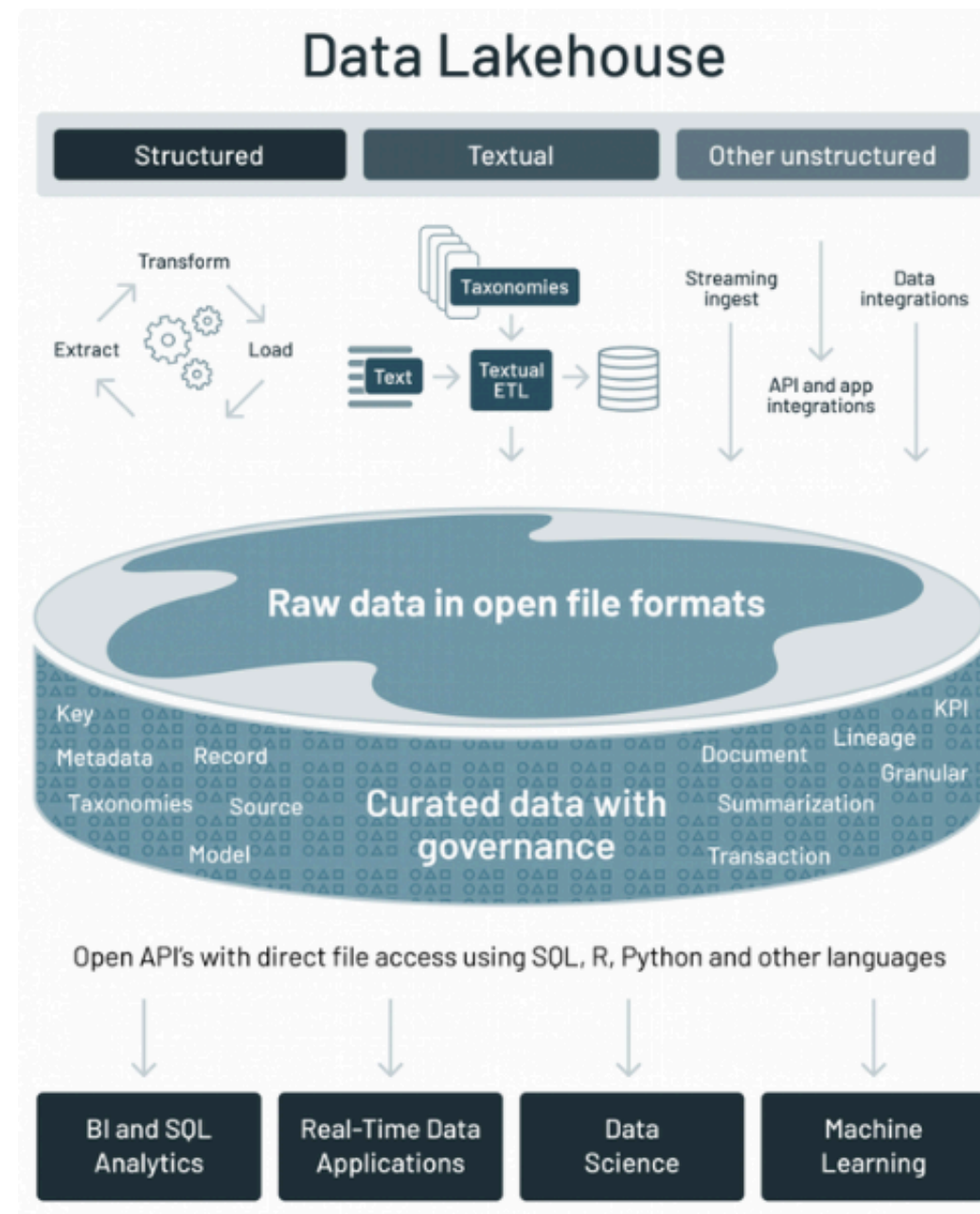
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Birth of the Lakehouse



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Birth of the Lakehouse

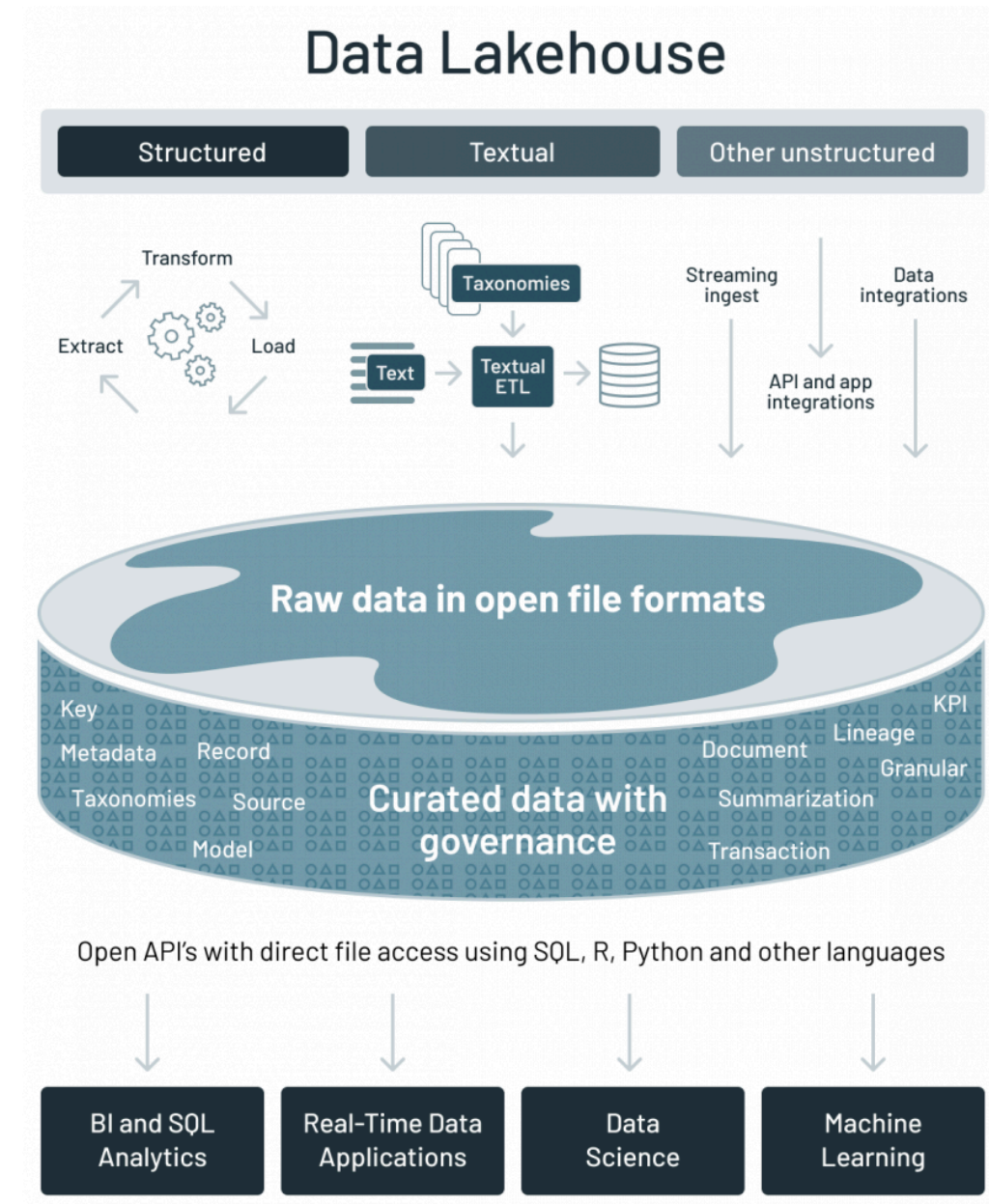


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The Databricks Lakehouse

The Databricks Lakehouse Platform

- Single platform for all data workloads
- Simplified architecture
- Collaborative environment



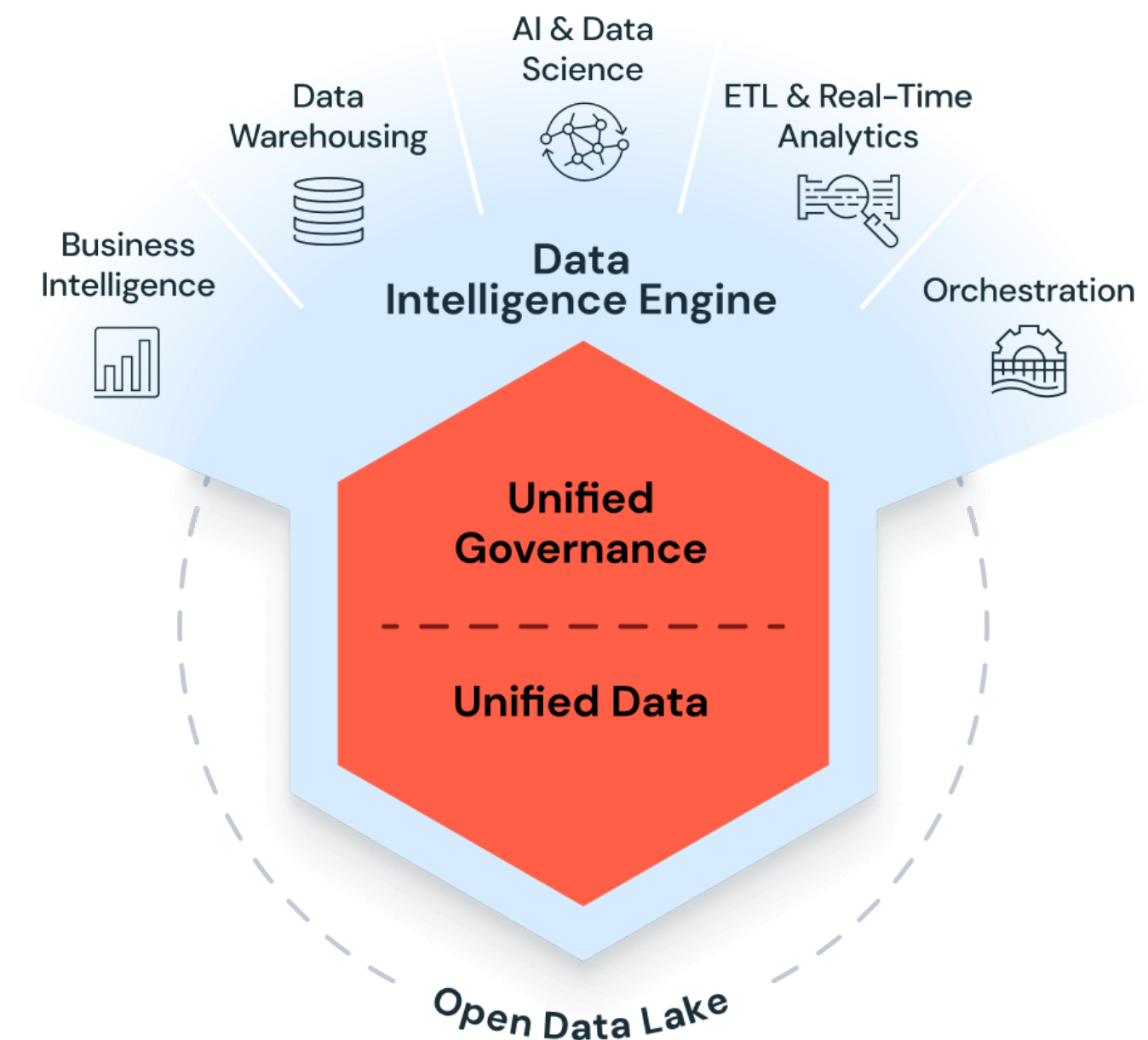
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The Databricks Data Intelligence Platform

Evolution from the Lakehouse

Databricks has expanded the Lakehouse vision to create the first *Data Intelligence Platform*

- Same core architecture of the Lakehouse
- Built-in AI
- First-class support for custom AI applications



¹ <https://www.databricks.com/product/data-intelligence-platform>

Databricks Architecture Benefits

Unification

- Every use case from AI to BI
- Benefits of data warehouse and data lake



Multi-Cloud

- Bring powerful platform to your data
- No lock-in to a specific cloud platform



Databricks Development Benefits

Collaborative

- Every data persona
- Ability to work in same platform in real-time



Open-Source

- Underpinned by *Apache Spark*
- Support for most popular languages (Python, R, Scala, SQL)



Let's practice!
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Setting up a Databricks workspace example

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Kevin Barlow
Data Practitioner

Let's practice!
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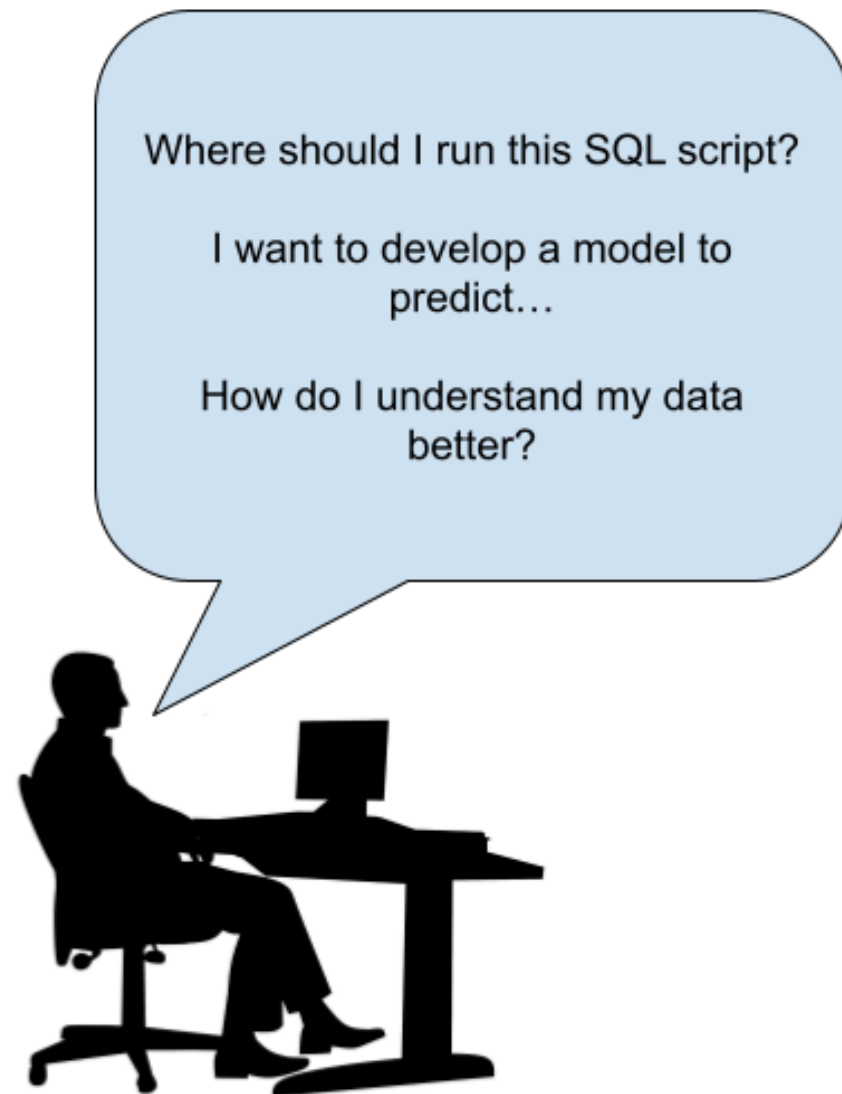
Databricks Architecture

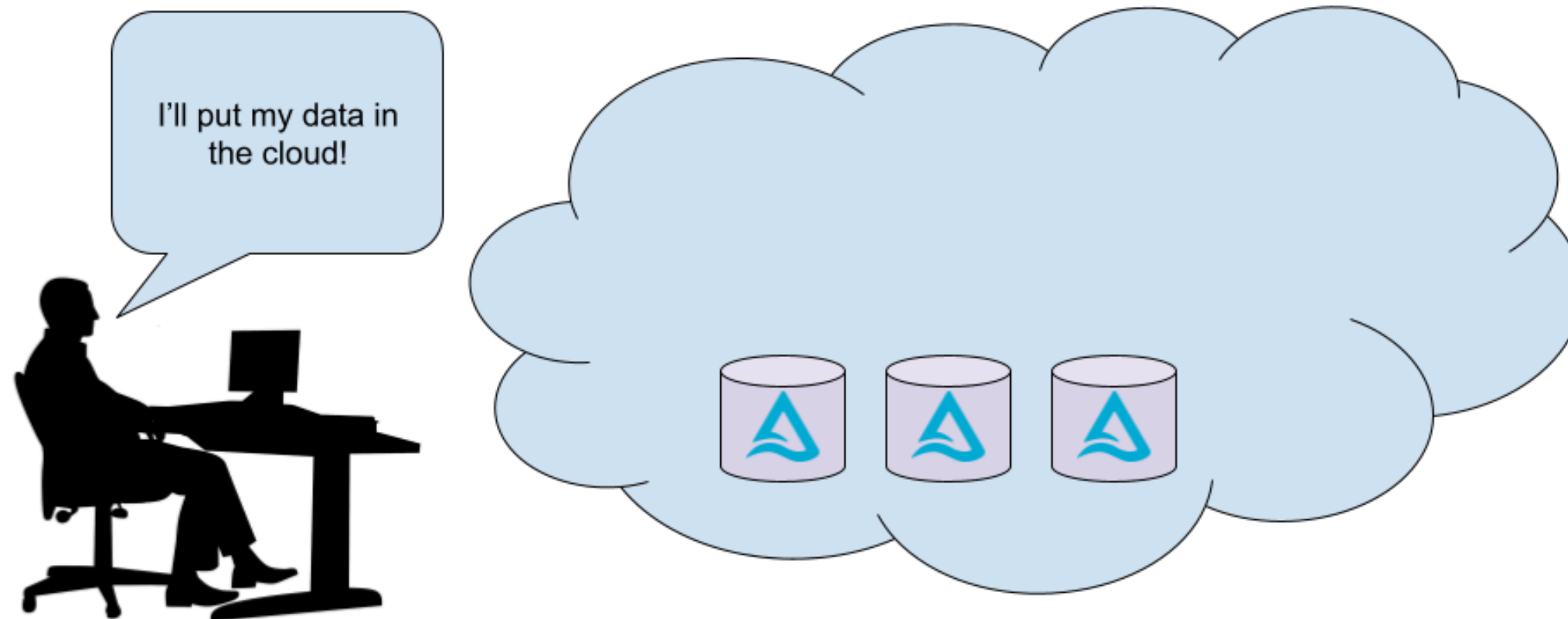
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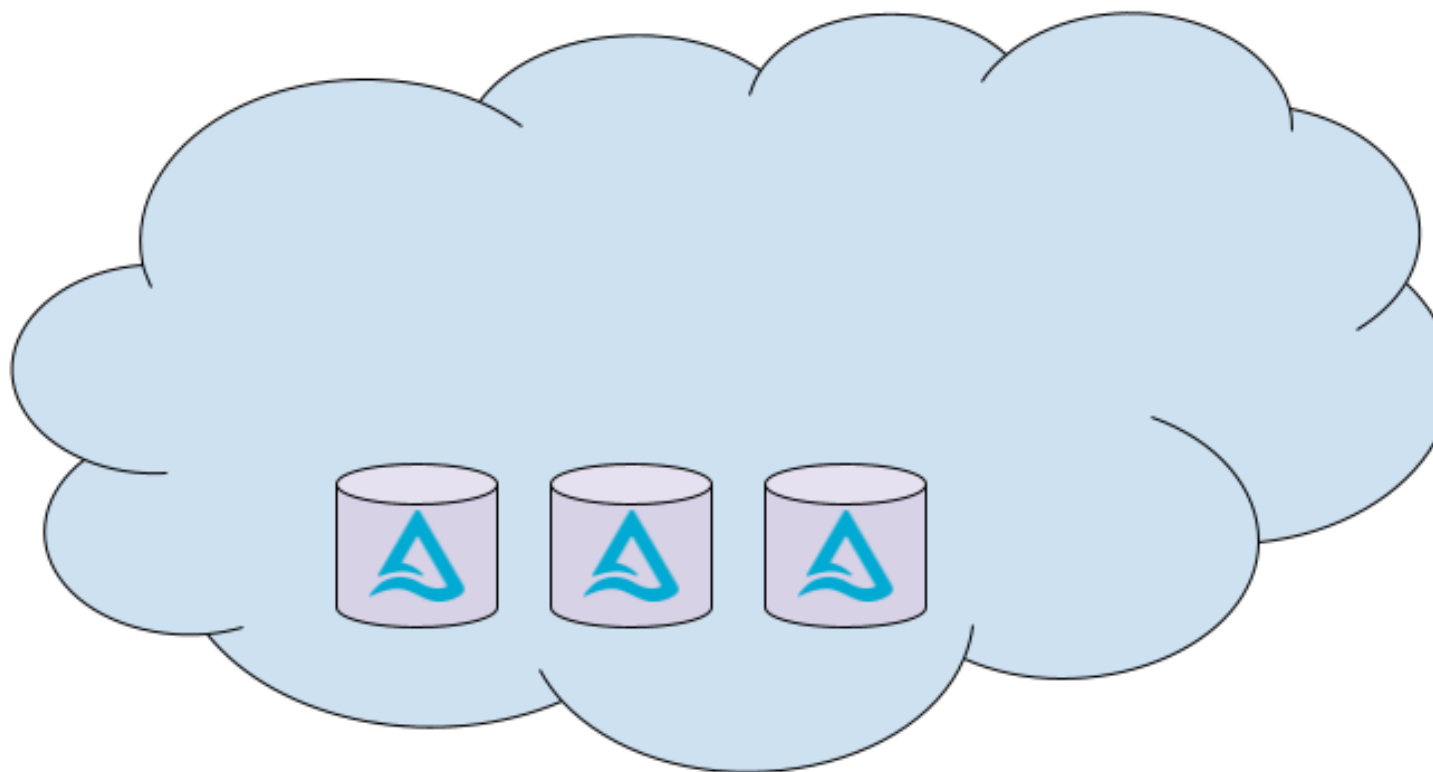
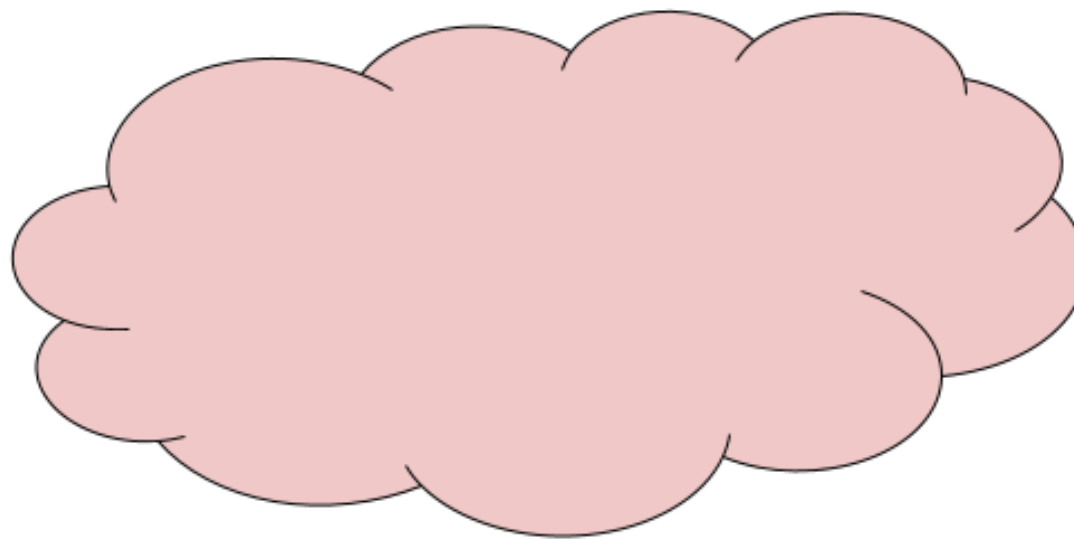


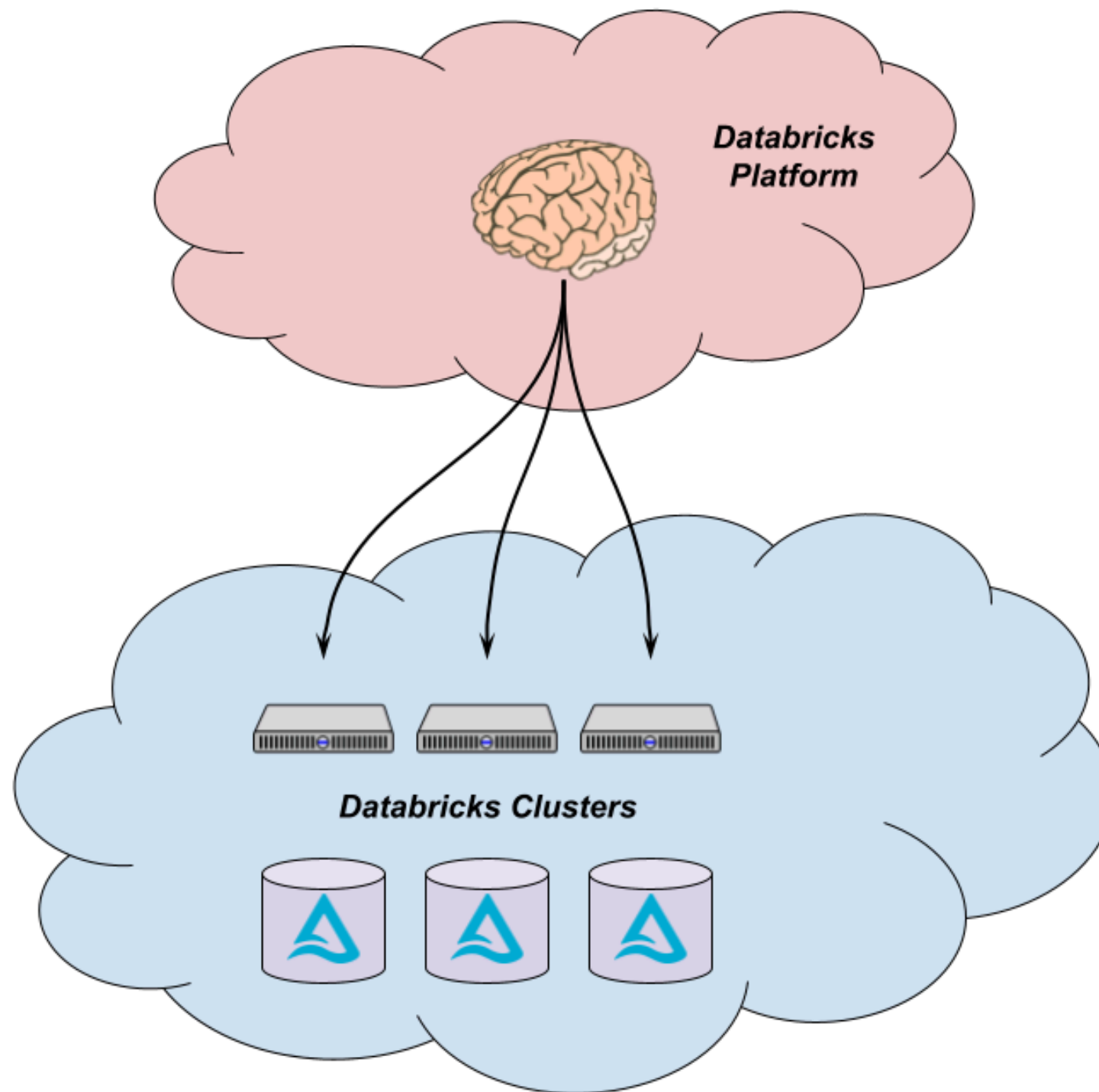
Kevin Barlow

Data Analytics Practitioner









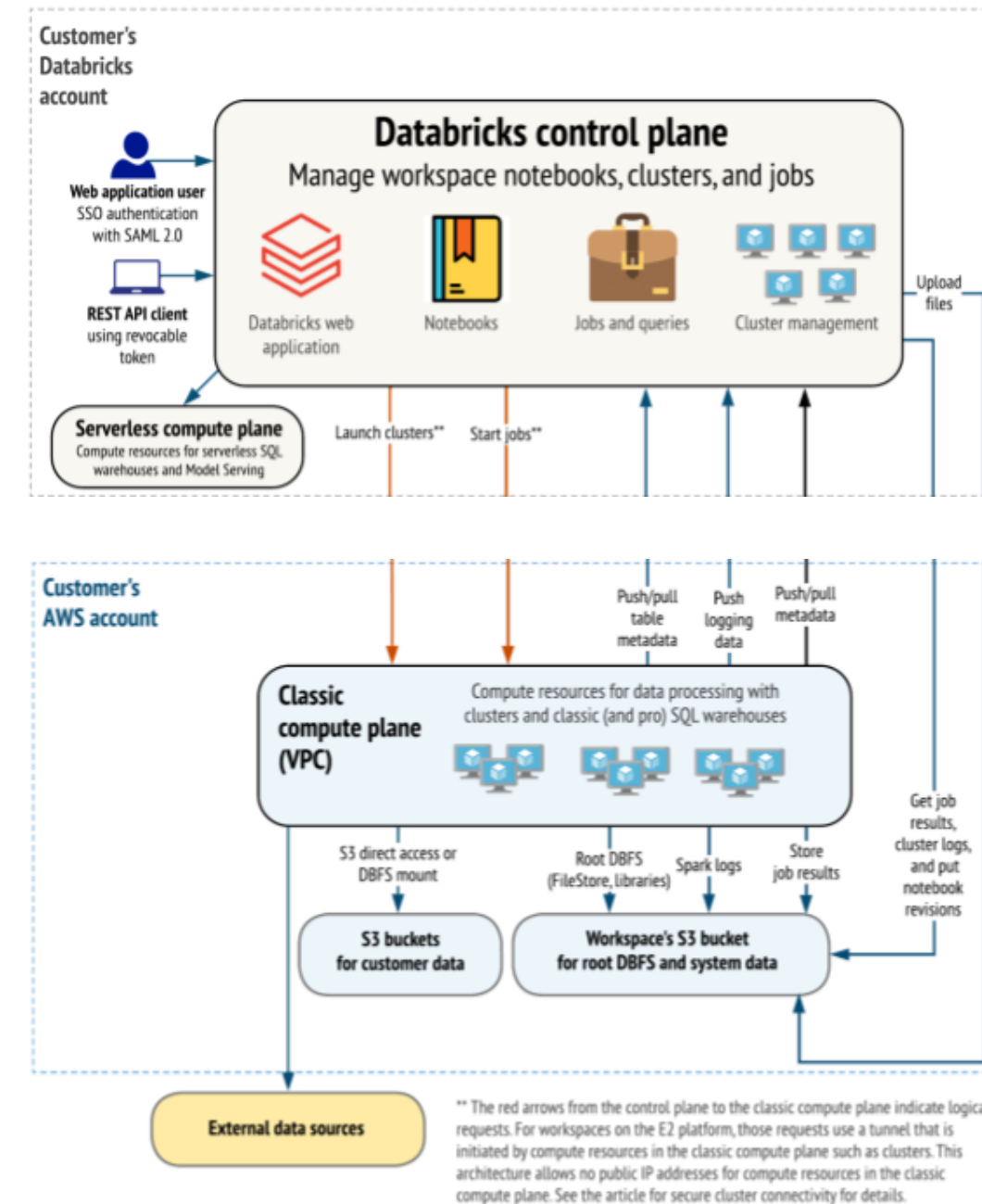
The Databricks Architecture

Control Plane

- Databricks owned environment in your cloud / region
- Hosts the UI, notebooks, and general code
- Orchestrates compute nodes for processing

Compute Plane

- Customer owned environment
- Location for data storage
- Customer networking, applications, etc.



Let's review!

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Administering a Databricks workspace

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Account Administrators

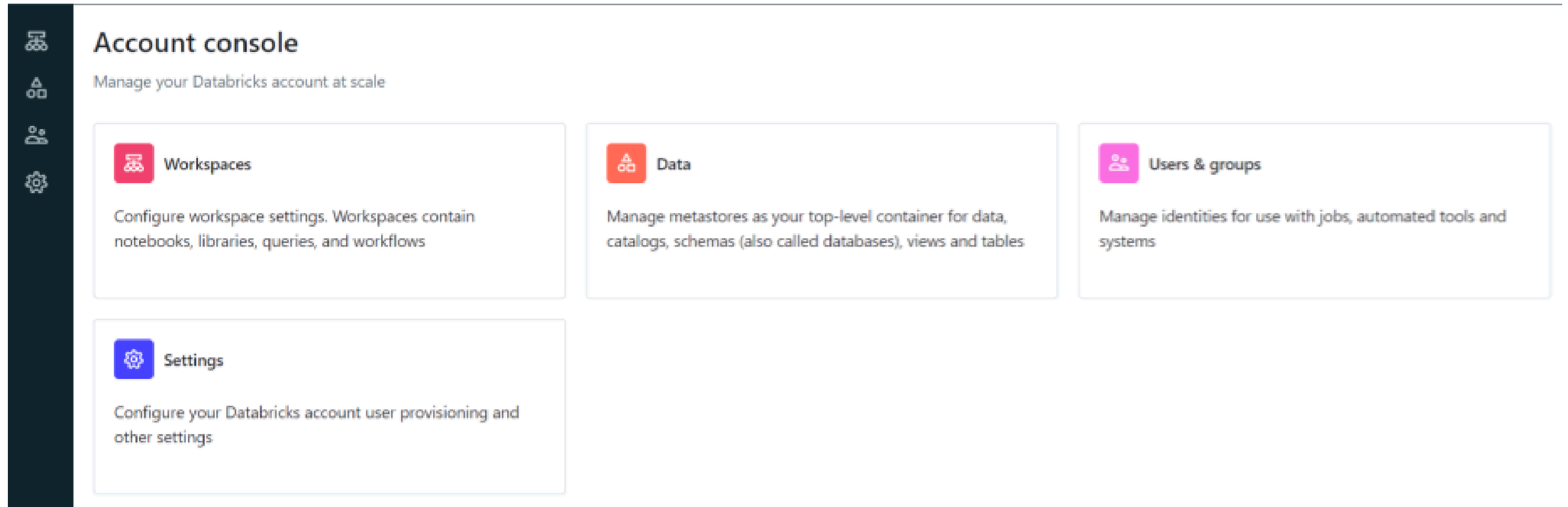
Key Responsibilities:

- Creating and managing workspaces
- Governing access to workspaces
- Managing the account subscription

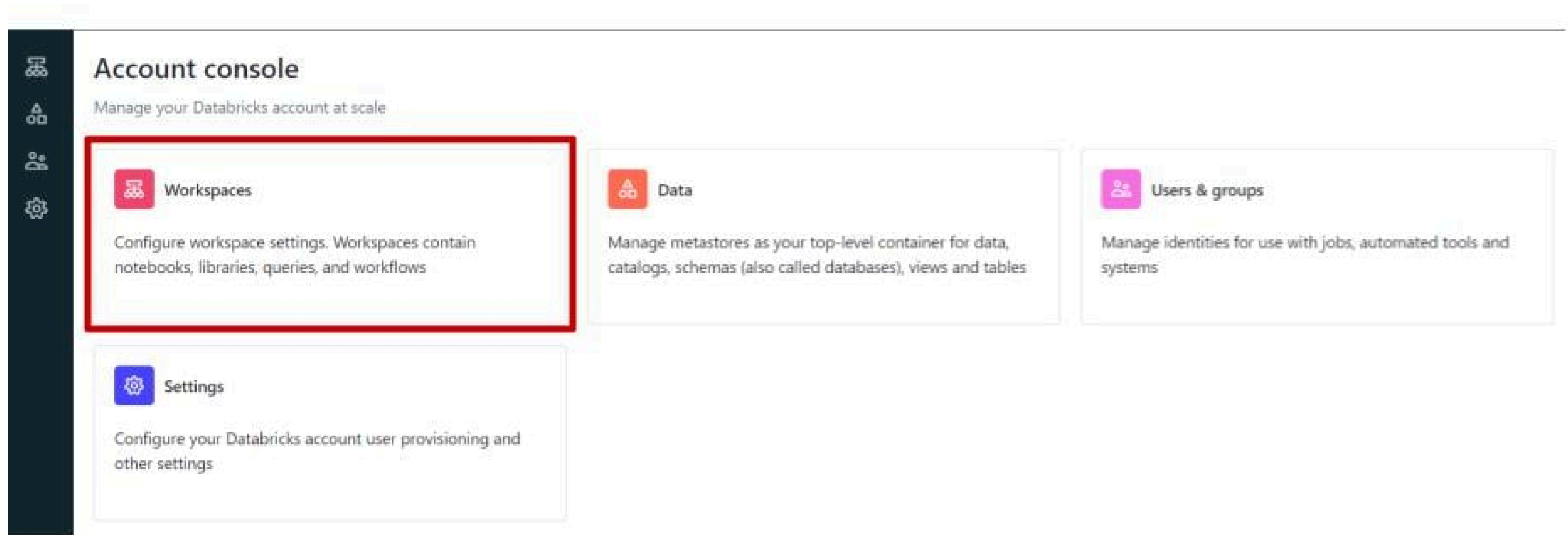


Account Console

<https://accounts.cloud.databricks.com/>

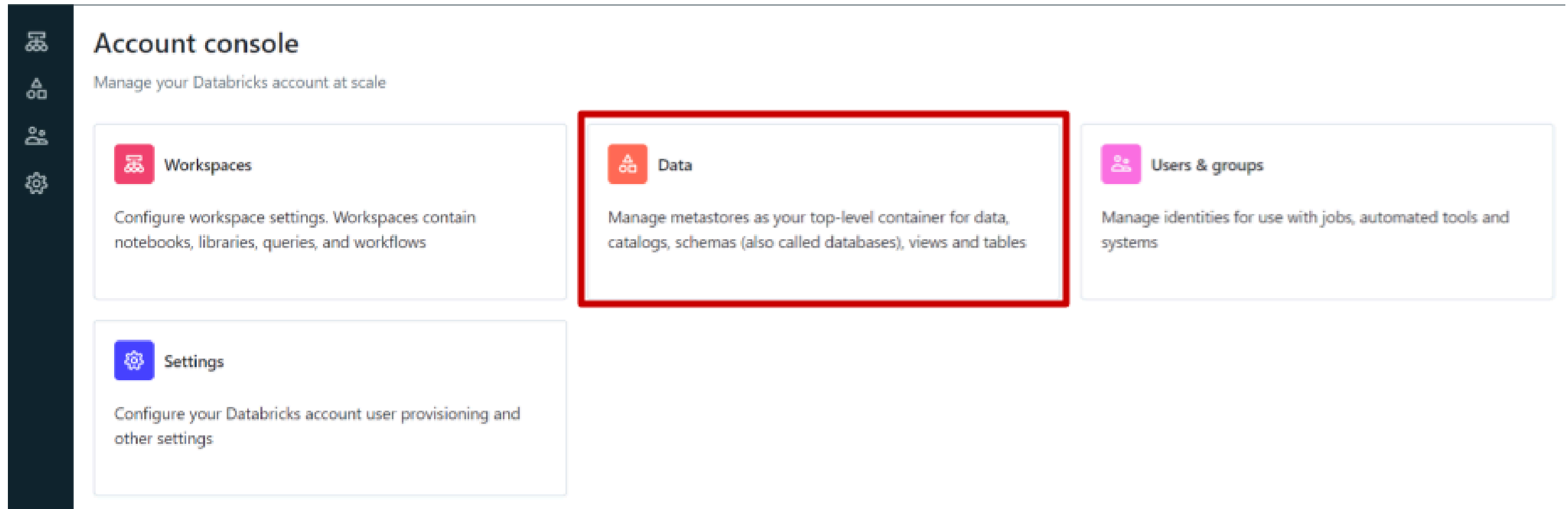


Workspaces



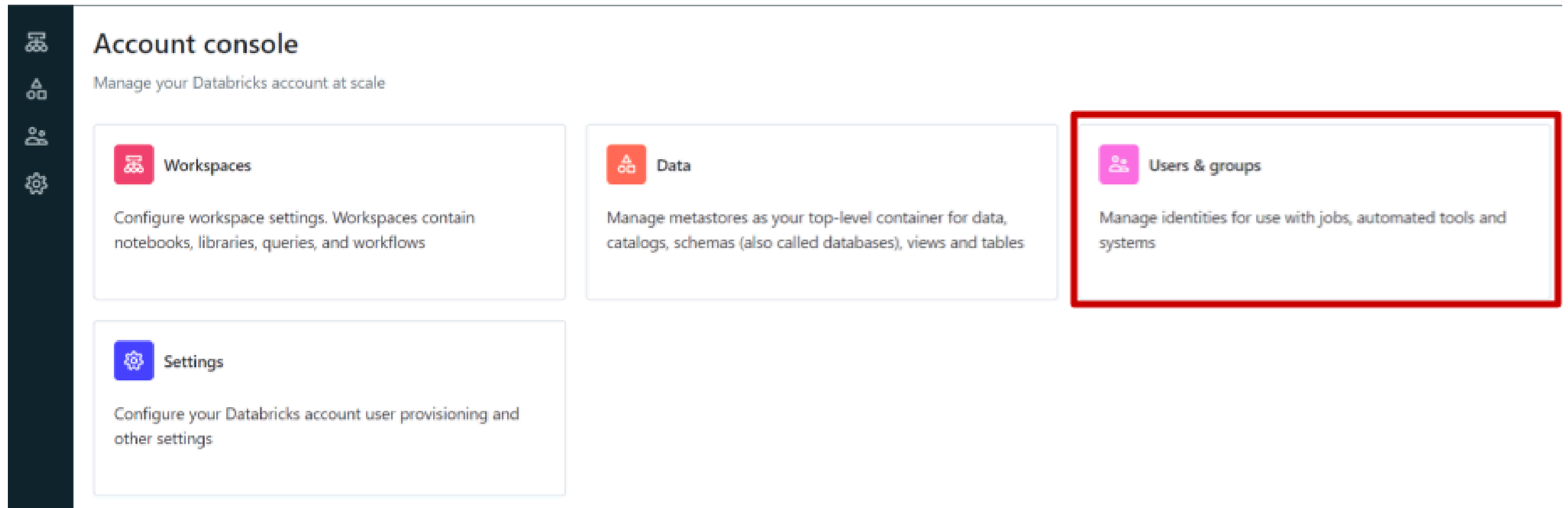
<https://accounts.cloud.databricks.com/>

Data



<https://accounts.cloud.databricks.com/>

Users & Groups



<https://accounts.cloud.databricks.com/>

Workspace Administrators

Key Responsibilities:

- Managing identities in your workspace
- Creating and managing compute resources

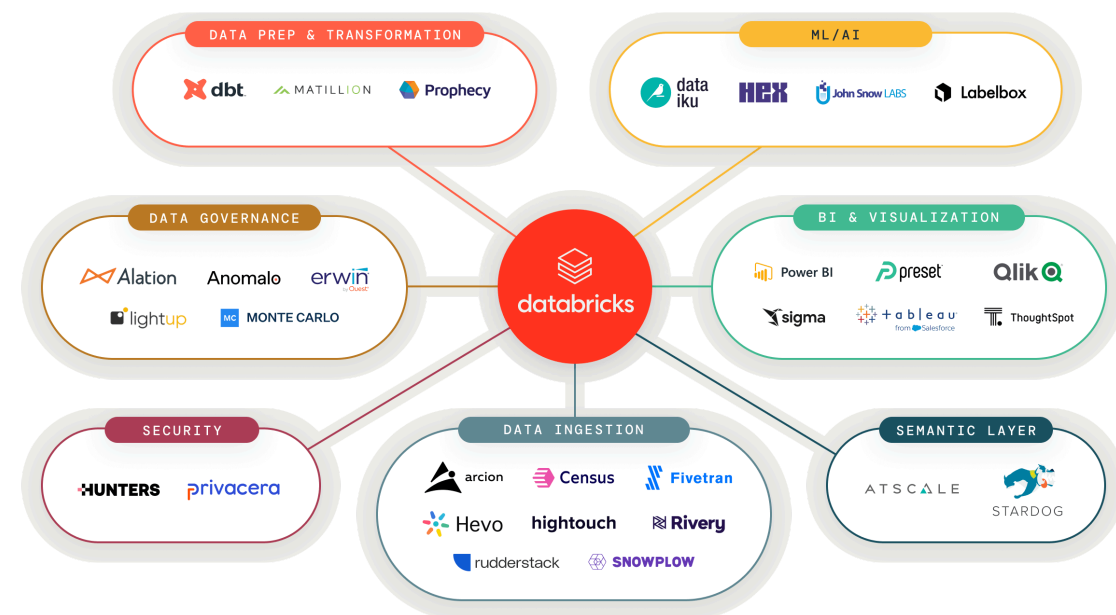
Admin Settings

Users Service principals Groups Global init scripts Workspace settings SQL settings Notification destinations SQL warehouse settings

Other Administrative Activities

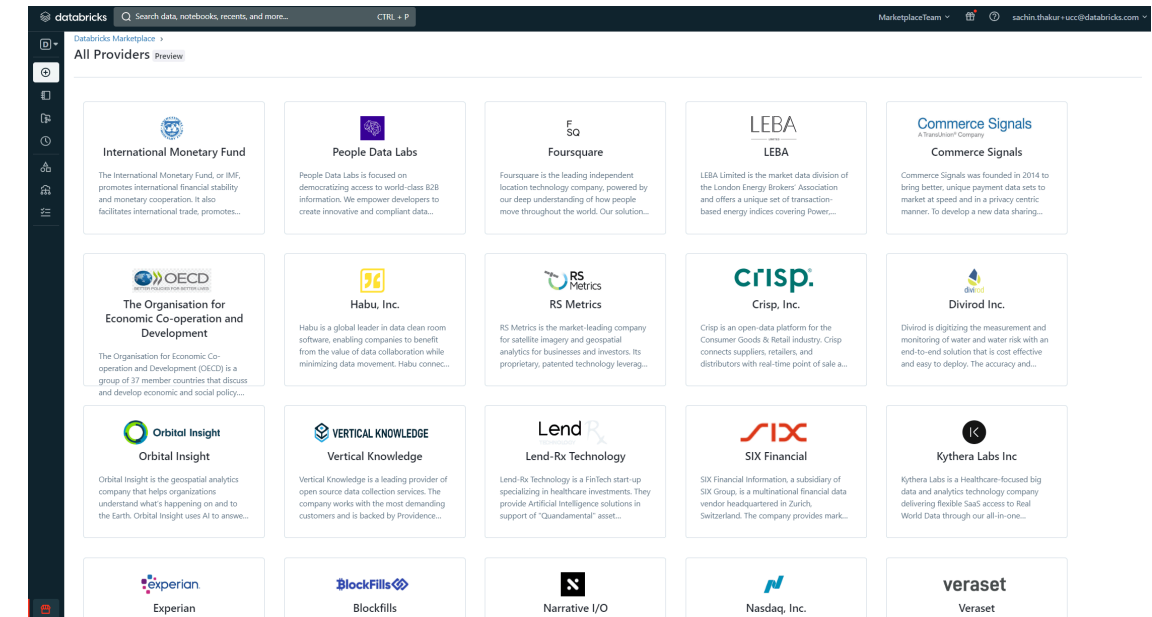
Partner Connect

- UI-based connection to partner technologies
- BI connections, ingestion tools, etc.



Databricks Marketplace

- Discover and access third-party datasets
- Integrate directly into your data catalogs



¹ <https://www.databricks.com/partnerconnect>

Let's review!

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